

**LM270WQ1**
Liquid Crystal Display**Product Specification****SPECIFICATION
FOR
APPROVAL**

(◆) Preliminary Specification
() Final Specification

Title		27.0" QHD TFT LCD	
BUYER	General	SUPPLIER	LG Display Co., Ltd.
MODEL		*MODEL	LM270WQ1
		SUFFIX	SDG1

*When you obtain standard approval,
please use the above model name without suffix

SIGNATURE	DATE
/	
/	
/	

Please return 1 copy for your confirmation
With your signature and comments.

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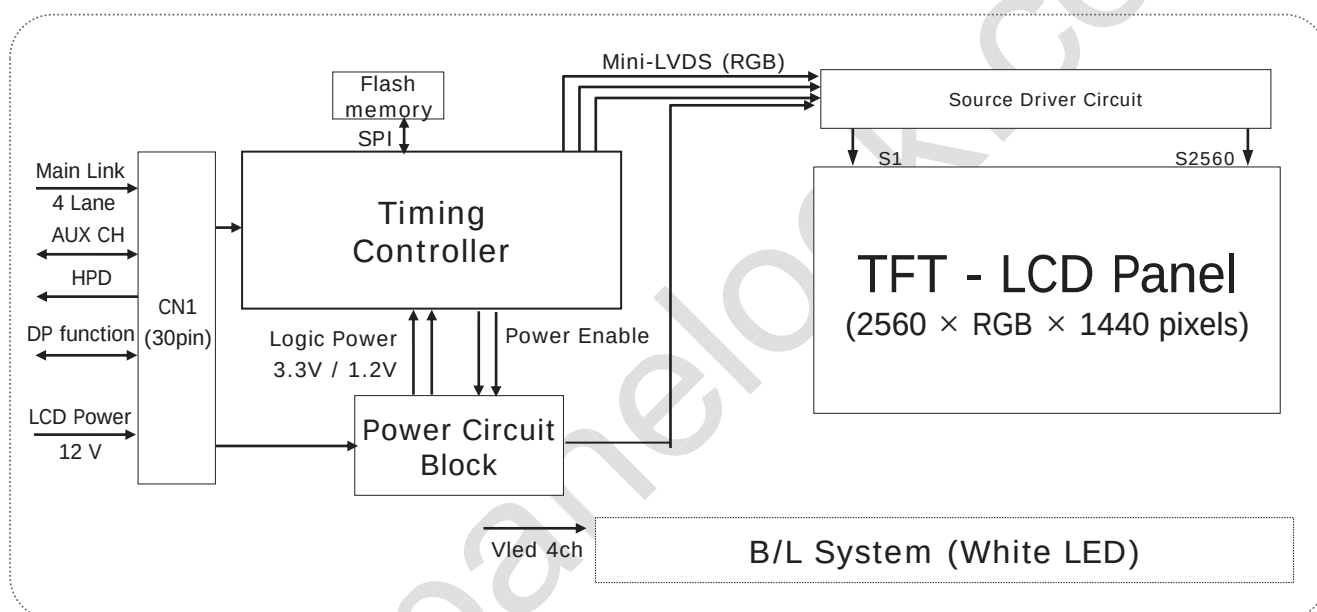


RECORD OF REVISIONS

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**LM270WQ1**
Liquid Crystal Display**Product Specification****1. General Description**

LM270WQ1 is a Color Active Matrix Liquid Crystal Display with Light Emitting Diode (White LED) backlight system without LED driver. The matrix employs a-Si Thin Film Transistor as the active element. It is a transmissive type display operating in the normally black mode. It has a 27inch diagonally measured active display area with QHD resolution (2560 vertical by 1440 horizontal pixel array). Each pixel is divided into Red, Green and Blue sub-pixels or dots which are arranged in vertical stripes. Gray scale or the brightness of the sub-pixel color is determined with a 8-bit gray scale signal for each dot, thus, presenting a palette of more than 16.7M colors. It has been designed to apply the 8-bit 4Lane Display Port interface. It is intended to support displays where high brightness, super wide viewing angle, high color saturation, and high color are important.

**General Features**

Active Screen Size	27.0 inches(68.47cm) diagonal
Outline Dimension	630.0(H) x 368.2(V) x 14.9(D) mm(Typ.)
Pixel Pitch	0.2331 mm x 0.2331 mm
Pixel Format	2560 horiz. By 1440 vert. Pixels RGB stripes arrangement
Color Depth	16.7 Million colors (8Bit)
Luminance, White	350 cd/m ² (Center 1Point, Typ.)
Viewing Angle(CR>10)	View Angle Free (R/L 178(Typ.), U/D 178(Typ.))
Power Consumption	Total 34.94 Watt (Typ.) (9.24 Watt @VLCD, 25.7 Watt w/o driver)
Weight	2,620g (Typ.)
Display Operating Mode	Transmissive mode, normally black
Surface Treatment	Hard coating(3H), Anti-glare treatment of the front polarizer



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2. Absolute Maximum Ratings

The following are maximum values which, if exceeded, may cause faulty operation or damage to the unit.

Table 1. ABSOLUTE MAXIMUM RATINGS

Parameter	Symbol	Values		Units	Notes
		Min	Max		
Power Input Voltage	V _{LCD}	-0.3	14	V _{dc}	at 25 ± 2°C
Operating Temperature	T _{OP}	0	50	°C	1, 2
Storage Temperature	T _{ST}	-20	60	°C	
Operating Ambient Humidity	H _{OP}	10	90	%RH	
Storage Humidity	H _{ST}	10	90	%RH	
LCM Surface Temperature (Operation)	T _{Surface}	0	65	°C	1,4

Note : 1. Temperature and relative humidity range are shown in the figure below.

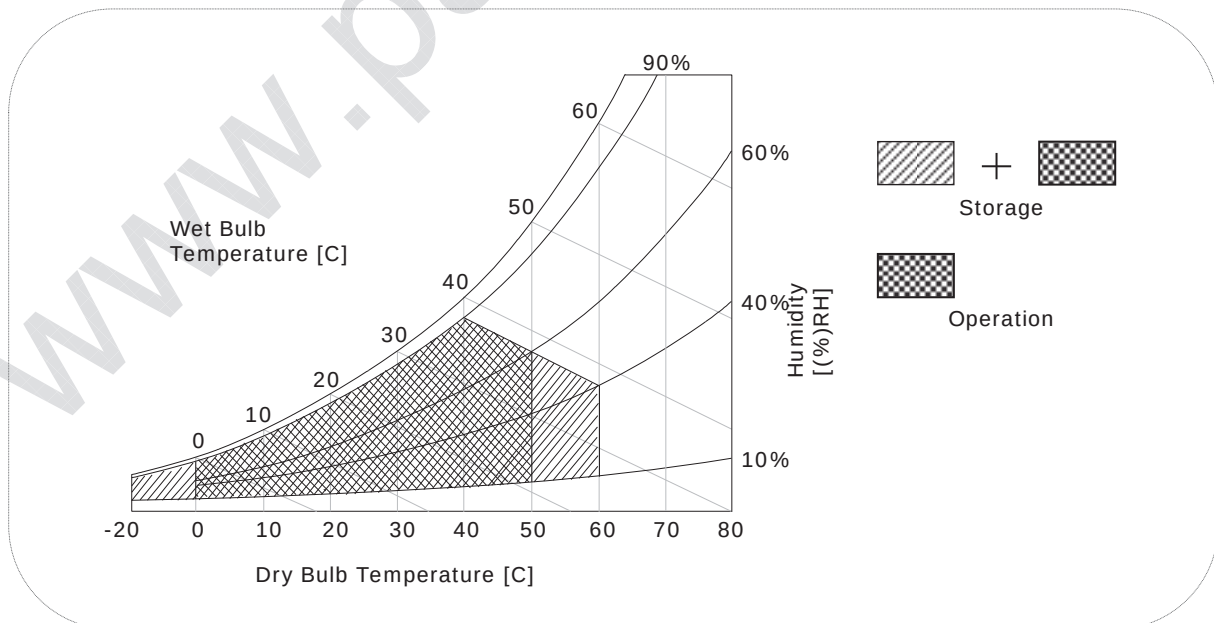
Wet bulb temperature should be 39 °C Max, and no condensation of water.

2. Maximum Storage Humidity is up to 40°C, 90% RH only for 4 corner light leakage Mura.

3. Storage condition is guaranteed under packing condition.

4. LCM Surface Temperature should be Min. 0 °C and Max. 65 °C under the V_{LCD}=12.0V, f_V=60Hz, 25 °C ambient Temperature no humidity control and LED string current is typical value.

FIG. 1 Temperature and relative humidity



Product Specification**3. Electrical Specifications****3-1-1. Electrical Characteristics**

It requires two power inputs. One is employed to power the LCD electronics and to drive the TFT array and liquid crystal. The second input power for the LED/Backlight, is typically generated by a LED Driver. The LED Driver is an external unit to the LCDs.

Table 2-1. ELECTRICAL CHARACTERISTICS

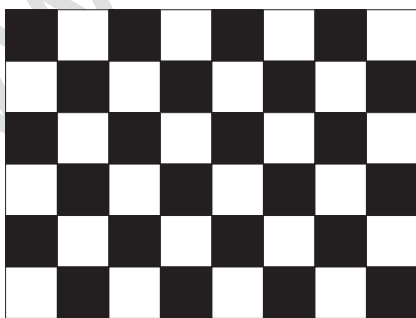
Parameter	Symbol	Values			Unit	Notes
		Min	Typ	Max		
MODULE :						
Power Supply Input voltage	V _{LCD}	11.6	12.0	12.4	Vdc	
Permissive Power Input Ripple	V _{dRF}	-		400	mVp-p	
Power Supply Input Current	I _{LCD-MOSAIC}	-	770	1000	mA	1
	I _{LCD-WHITE}	-	1050	1365	mA	2
Power Consumption	P _{LCD-MOSAIC}	-	9.24	12.0	Watt	1
	P _{LCD-WHITE}		12.6	16.38	Watt	2
Rush Current	I _{RUSH}	-	-	3.0	A	3

Note :

1. The specified current and power consumption are under the $V_{LCD}=12.0V$, $25 \pm 2^{\circ}C$, $f_V=60Hz$ condition whereas mosaic pattern(8 x 6) is displayed and f_V is the frame frequency.
2. The current is specified at the maximum current pattern.
3. The duration of rush current is about 2ms and rising time of power Input is 1ms(min.).

FIG.2 Pattern for Electrical characteristicspower consumption measurement

White : 255Gray
Black : 0Gray



Mosaic Pattern(8 x 6)

power input ripple

Maximum current pattern



White Pattern



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Table 2-2. LED BAR ELECTRICAL CHARACTERISTICS

Parameter	Symbol	Values			Unit	Notes
		Min.	Typ.	Max.		
LED String Current	Is	10	120	125	mA	1,2,5
LED String Voltage	Vs	50.2	53.6	57.0	V	1,5
Power Consumption	PBar	-	25.7	27.3	Watt	1,2,4
LED Life Time	LED_LT	30,000	-	-	Hrs	3

Notes) The LED Bar consists of 68 LED packages, 4 strings (parallel) x 17 packages (serial)

LED driver design guide

: The design of the LED driver must have specifications for the LED in LCD Assembly.

The performance of the LED in LCM, for example life time or brightness, is extremely influenced by the characteristics of the LED driver.

So all the parameters of an LED driver should be carefully designed and output current should be Constant current control.

Please control feedback current of each string individually to compensate the current variation among the strings of LEDs.

When you design or order the LED driver, please make sure unwanted lighting caused by the mismatch of the LED and the LED driver (no lighting, flicker, etc) never occurs.

When you confirm it, the LCD module should be operated in the same condition as installed in your instrument.

1. Specified values are for a single LED bar.
2. The specified current is defined as the input current for a single LED string with 100% duty cycle.
3. The LED life time is defined as the time when brightness of LED packages become 50% or less than the initial value under the conditions at $T_a = 25 \pm 2^\circ\text{C}$ and LED string current is typical value.
4. The power consumption shown above does not include loss of external driver.
The typical power consumption is calculated as $P_{\text{Bar}} = V_s(\text{Typ.}) \times I_s(\text{Typ.}) \times \text{No. of strings}$.
The maximum power consumption is calculated as $P_{\text{Bar}} = V_s(\text{Max.}) \times I_s(\text{Typ.}) \times \text{No. of strings}$.
5. LED operating conditions are must not exceed Max. ratings.

**LM270WQ1**
Liquid Crystal Display**Product Specification****3-2. Interface Connections**

This LCD module employs 30pin connectors are used for the module electronics and 6pin connectors are used for the integral backlight system.

3-2-1. LCD Module (CN1)

-LCD Connector(CN1). : FI-X30SSL-HF (manufactured by JAE)

The pin configuration for the 30 pin connector is shown in the table below.

Table 3-1. MODULE CONNECTOR(CN1) PIN CONFIGURATION

Pin No.	Symbol	Description	Pin No.	Symbol	Description
1	DDC_SCL	DDC for Clock	16	Lane 3P	True Signal for Main Link 3
2	DDC_SDA	DDC for Data	17	Lane 3N	Component Signal for Main Link 3
3	GND	High Speed Ground for Auxiliary Channel	18	GND	High Speed Ground
4	AUX_CH N	Component Signal for Auxiliary Channel	19	SPDIF	Audio Output from DP RX
5	AUX_CH P	True Signal for Auxiliary Channel	20	VIDEO_ON	Video Status from DP RX
6	GND	High Speed Ground for Main Link0	21	HPD	Hot Plug Detect Signal
7	Lane 0P	True Signal for Main Link 0	22	VSYNC	Vertical Sync Output from DP RX
8	Lane 0N	Component Signal for Main Link 0	23	GND	GND for Main Power
9	GND	High Speed Ground for Main Link0	24	GND	GND for Main Power
10	Lane 1P	True Signal for Main Link 1	25	GND	GND for Main Power
11	Lane 1N	Component Signal for Main Link 1	26	VLCD	12V for LCM Main Power
12	GND	High Speed Ground for Main Link1	27	VLCD	12V for LCM Main Power
13	Lane 2P	True Signal for Main Link 2	28	VLDC	12V for LCM Main Power
14	Lane 2N	Component Signal for Main Link 2	29	VLDC	12V for LCM Main Power
15	GND	High Speed Ground for Main Link3	30	NC	Not Connection

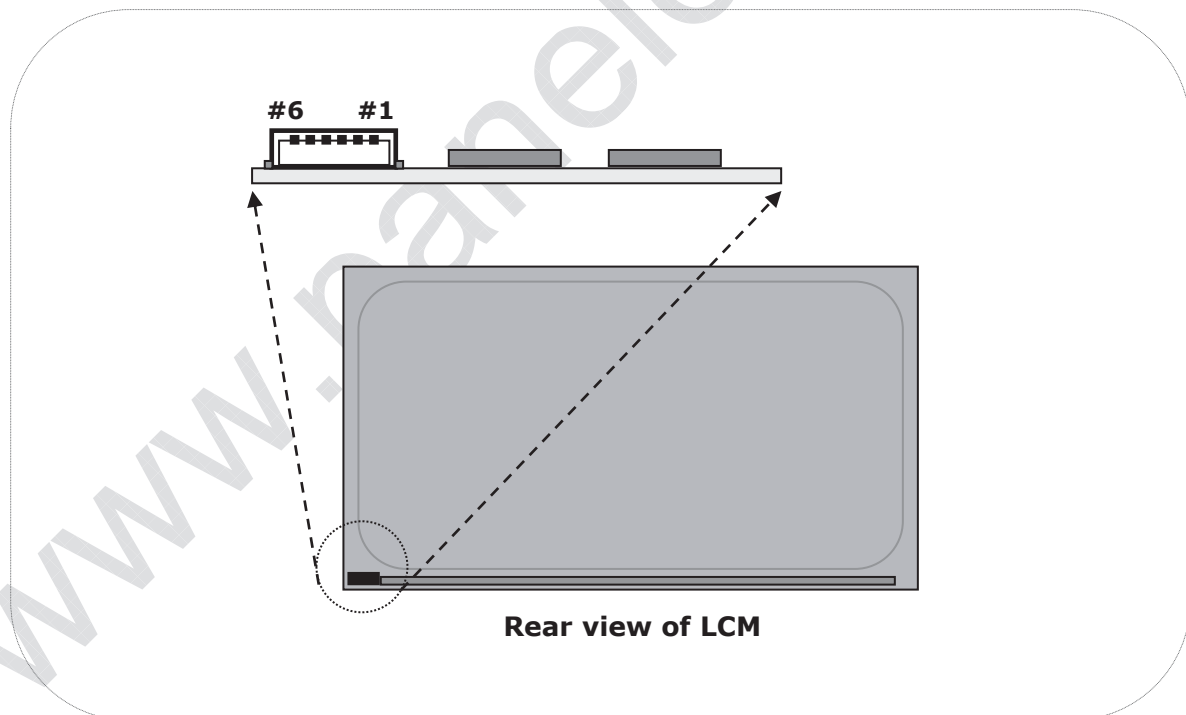
- Notes : 1. All GND(ground) pins should be connected together to the LCD module's metal frame.
2. All VLCD (power input) pins should be connected together.
3. Connector(Receptacle) : FI-X30SSL-HF(JAE) or 20389-Y30E-01(I-PEX)
Mating Connector(Plug) : FI-X30HL(JAE) or 20385-Y30T-12F(I-PEX)

Product Specification**3-2-2. BACKLIGHT CONNECTOR PIN CONFIGURATION(CN2)**

The LED interface connector is a model SM06B-SHJH(HF), wire-locking type manufactured by JST. The mating connector is a SHJP-06V-S(HF) or SHJP-06-A-K(HF) and Equivalent. The pin configuration for the connector is shown in the table below.

Table 3-2. LED CONNECTOR PIN CONFIGURATION

Pin	Symbol	Description
1	FB1	Channel1 Current Feedback
2	FB2	Channel2 Current Feedback
3	VLED	LED Power Supply
4	VLED	LED Power Supply
5	FB3	Channel3 Current Feedback
6	FB4	Channel4 Current Feedback

**Figure 3. Backlight connector view**



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3-3. Signal Timing Specifications

This is the signal timing required at the input of the User connector. All of the interface signal timing should be satisfied with the following specifications for it's proper operation.

Table 4. TIMING TABLE for Frame Rate 60Hz

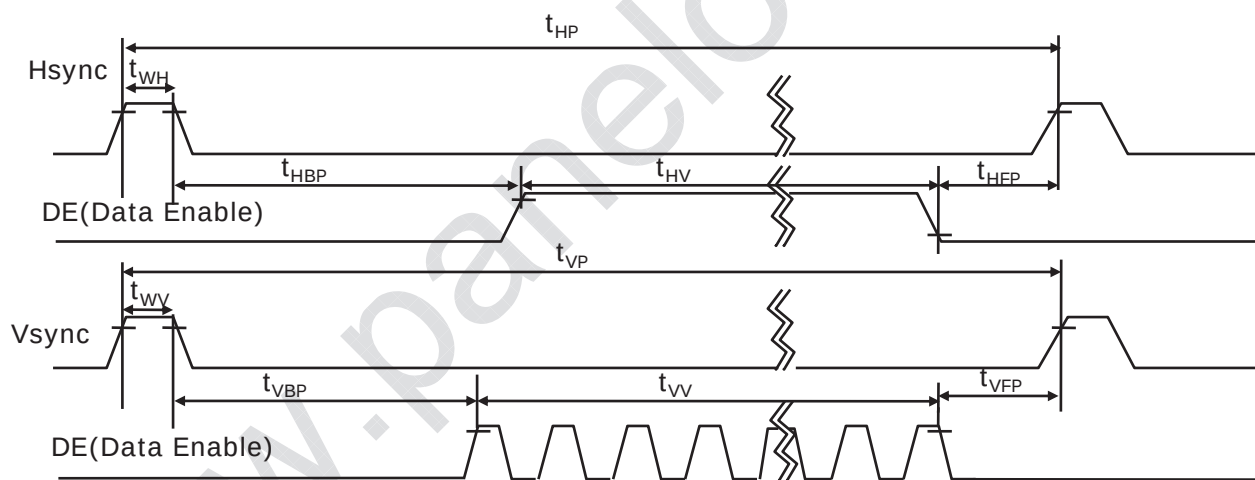
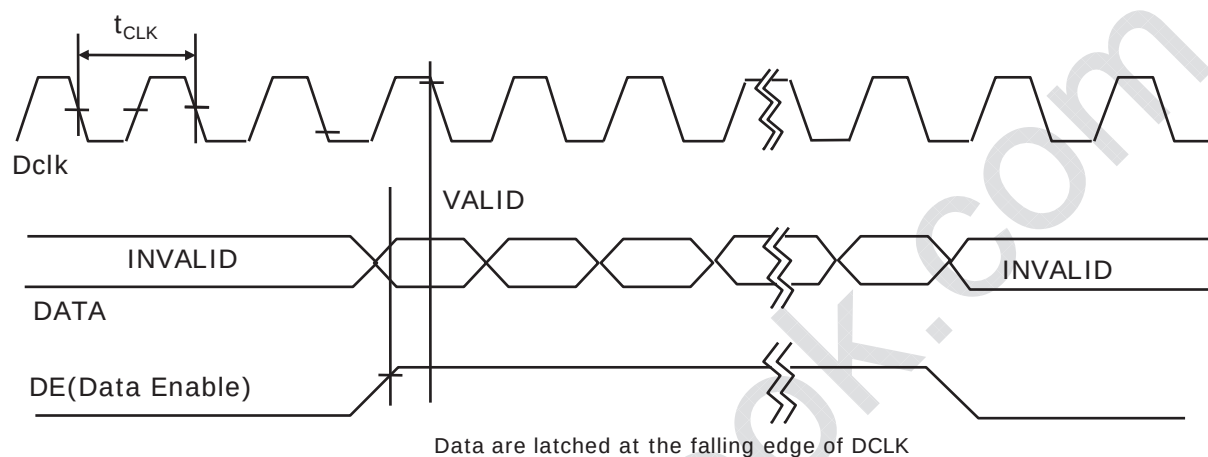
ITEM		SYMBOL	Min	Typ	Max	Unit	Note
DCLK	Period	tCLK	4.14	4.14	4.14	ns	
	Frequency	fCLK	241.5	241.5	241.5	MHz	-
Hsync	Period	tHP	2720	2720	2720	tCLK	
	Width-Active	tWH	32	32	32		
Vsync	Period	tVP	1481	1481	1481	tHP	
	Frequency	fV	59.95	59.95	59.95	Hz	
	Width-Active	tWV	5	5	5	tHP	
Data Enable	Horizontal Valid	tHV	2560	2560	2560	tCLK	
	Horizontal Back Porch	tHBP	80	80	80		
	Horizontal Front Porch	tHFP	48	48	48		
	Horizontal Blank	-	160	160	160		tWH+ tHBP+ tHFP
	Vertical Valid	tVV	1440	1440	1440	tHP	
	Vertical Back Porch	tVBP	33	33	33		
	Vertical Front Porch	tVFP	3	3	3		
	Vertical Blank	-	41	41	41		tWV+ tVBP+ tVFP

Note: Hsync period and Hsync width-active should be even number times of tCLK.

If the value is odd number times of tCLK, display control signal can be asynchronous.

In order to operate this LCM a Hsync, Vsync, and DE(data enable) signals should be used.

1. The performance of the electro-optical characteristics may be influenced by variance of the vertical refresh rates.
2. Vsync and Hsync should be keep the above specification.
3. Hsync Period, Hsync Width, and Horizontal Back Porch should be any times of of character number(8).
4. The polarity of Hsync, Vsync is not restricted.

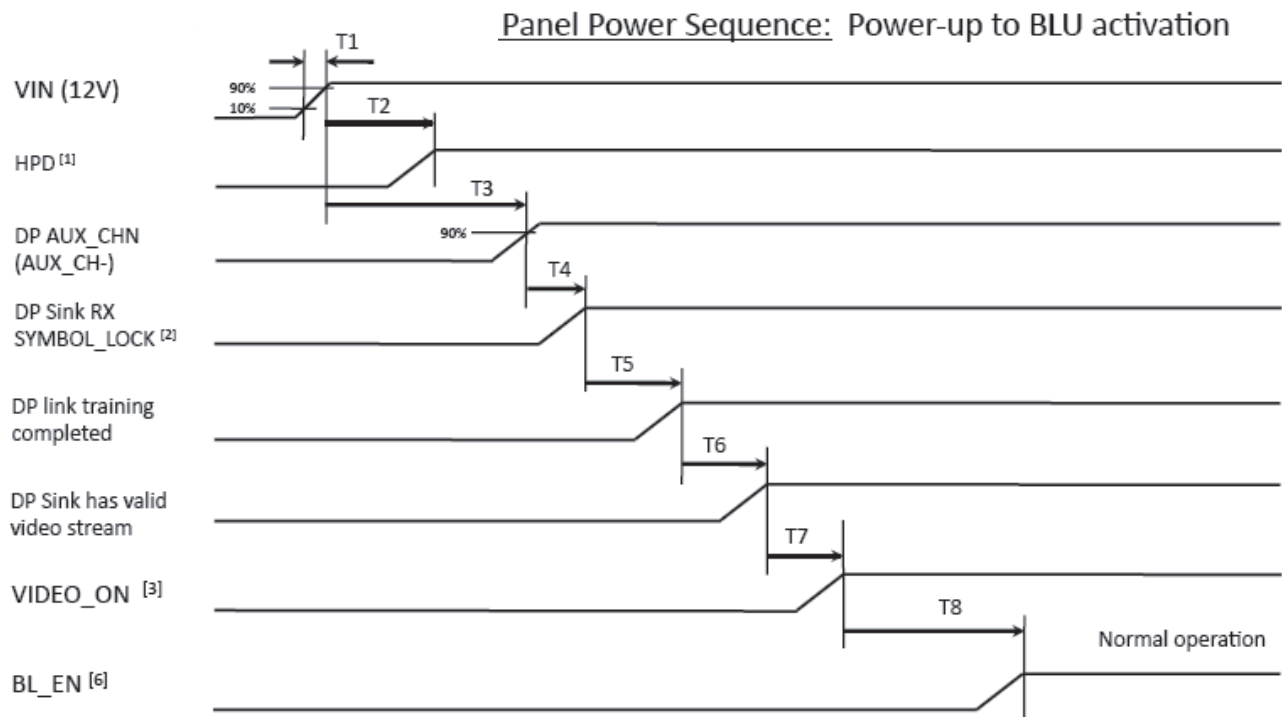
Product Specification**3-4. Signal Timing Waveforms**


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3-5. Color Data Reference

The Brightness of each primary color(red,green,blue) is based on the 8-bit gray scale data input for the color; the higher the binary input, the brighter the color. The table below provides a reference for color versus data input.

Table 5. COLOR DATA REFERENCE

Color		Input Color Data																							
		RED								GREEN								BLUE							
		MSB							LSB	MSB							LSB	MSB							LSB
		R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
Basic Color	Black	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Red (255)	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Green (255)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	Blue (255)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Cyan	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Magenta	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
	Yellow	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
	White	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RED	RED (000) Dark	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	RED (001)	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	...																								
	RED (254)	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	RED (255)	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GREEN	GREEN (000) Dark	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	GREEN (001)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
	...																								
	GREEN (254)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
	GREEN (255)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
BLUE	BLUE (000)Dark	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	BLUE (001)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
	...																								
	BLUE (254)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	0
	BLUE (255)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

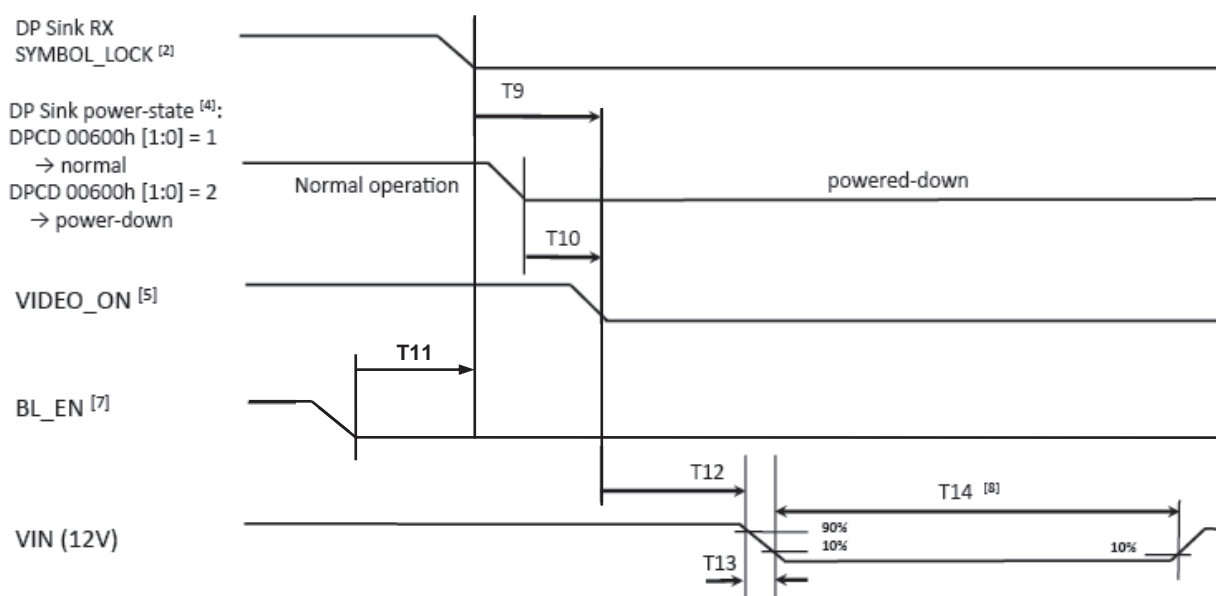
**LM270WQ1**
Liquid Crystal Display**Product Specification****3-6. Power Sequence & Dip condition for LCD Module****3-6-1. Power Sequence**

Parameter	Min	Max	Units
T1	0.5	10	ms
T2	0	-	ms
T3	-	-	ms
T4	0	-	ms
T5	0	-	ms

Parameter	Min	Max	Units
T6	0	-	ms
T7	0	200	ms
T8	500	-	ms

Notes : [1] HPD is asserted high by Sink at power-up
[2] SYMBOL_LOCK indicated by contents of Sink DPCD registers 00202h to 00205h
[3] VIDEO_ON asserted high by Sink when video to panel is valid
[6] BL_EN is an active-high MLB enable signal for panel BLU

- Notes : 1. Please avoid floating state of interface signal at invalid period.
2. When the interface signal is invalid, be sure to pull down the power supply for LCD V_{LCD} to 0V.
3. LED power must be turn on after power supply for LCD and interface signal are valid.
4. Be sure to follow Power sequence at these case
(① AC/DC Power On/Off, ② Resolution change, ③ Color depth change, etc.)

**LM270WQ1**
Liquid Crystal Display**Product Specification****3-6. Power Sequence & Dip condition for LCD Module****3-6-2. Power Sequence**Panel Power Sequence: BLU de-activation, power-down

Parameter	Min	Max	Units
T9	0	10	ms
T10	0	5	ms
T11	200	-	ms
T12	0	-	ms
T13	0.01	1000	ms
T14	1000	-	ms

Notes : [2] SYMBOL_LOCK indicated by contents of Sink DPCD registers 00202h to 00205h

[4] Power-state set by Source in Sink DPCD register 00600h

[5] VIDEO_ON asserted low by Sink because of :

- 1) loss of SYMBOL_LOCK or
- 2) DP Sink is powered down

[7] BL_EN must be asserted low by system as rapidly as possible when video is invalid to avoid visible artifacts

[8] T14 defines minimum off-time for 12V power

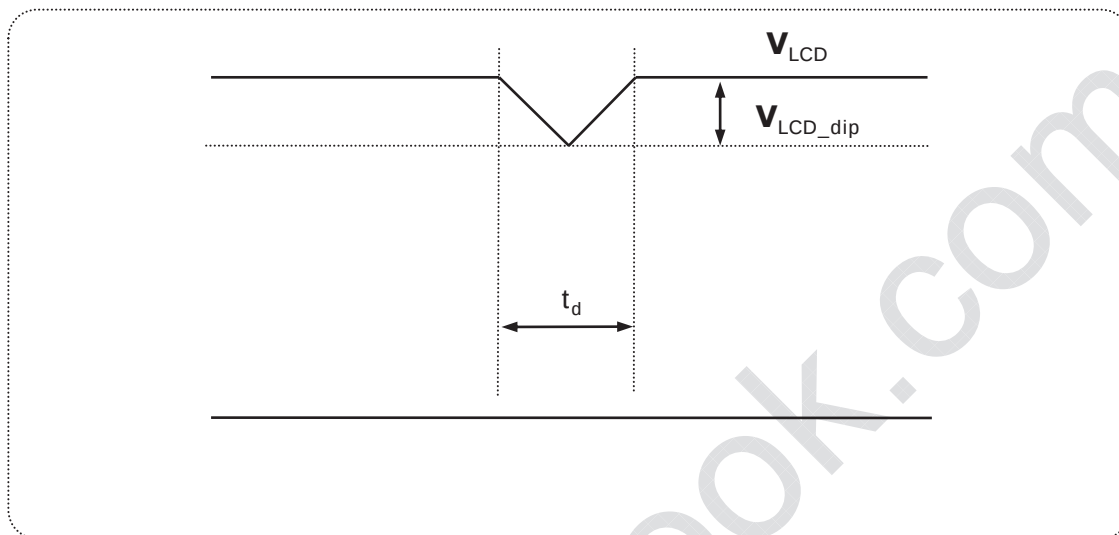
[9] min. times of 0 indicate precedence ordering of events, e.g. where actual timing is TBD



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3-6-3. VLCD Power Dip Condition



Notes :

Dip condition

$$V_{LCD_dip} \leq V_{LCD_typ} \times 0.2, \quad t_d \leq 10ms$$

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4. Optical Specifications

Optical characteristics are determined after the unit has been 'ON' for approximately 30 minutes in a dark environment at $25 \pm 2^\circ\text{C}$. The values specified are at an approximate distance 50cm from the LCD surface at a viewing angle of Φ and θ equal to 0° and aperture 1 degree.

FIG. 4 presents additional information concerning the measurement equipment and method.

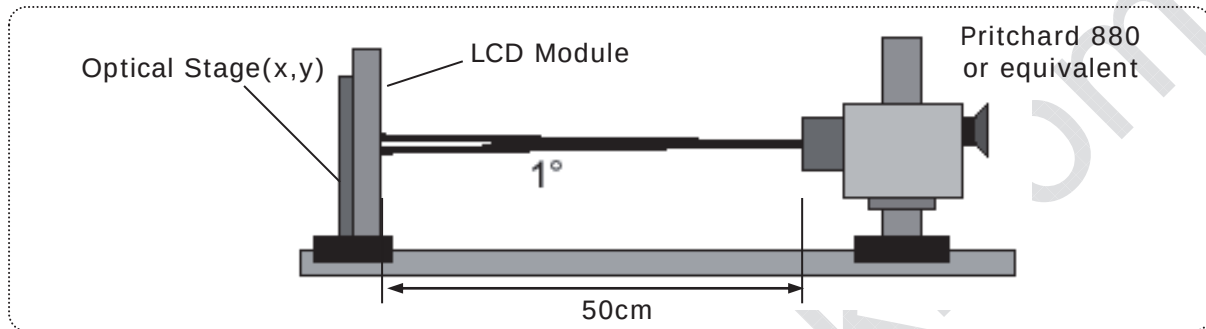


FIG. 4 Optical Characteristic Measurement Equipment and Method

Table 6. OPTICAL CHARACTERISTICS ($T_a = 25^\circ\text{C}$, $V_{\text{LCD}} = 12.0\text{V}$, $f_v = 60\text{Hz}$ Dclk=241.5MHz, $I_{\text{BL}} = 120\text{mA}$)

Parameter		Symbol	Values			Units	Notes
			Min	Typ	Max		
Contrast Ratio		CR	700	1000	-		1
Surface Luminance, white		L_{WH}	280	350	-	cd/m^2	2
Luminance Variation		δ_{WHITE}	75			%	3
Response Time	Gray to Gray	T_{GTG}	-	14	25	ms	5
Color Coordinates [CIE1931]	RED	R_x	Typ -0.03	0.653	Typ +0.03		
		R_y		0.336			
	GREEN	G_x		0.295			
		G_y		0.640			
	BLUE	B_x		0.146			
		B_y		0.042			
	WHITE	W_x		0.313			
		W_y		0.329			
Color Shift	Horizontal	$\theta_{\text{CST_H}}$	-	178	-	Degree	6
	Vertical	$\theta_{\text{CST_V}}$	-	178	-		
Viewing Angle (CR>10)							
General	Horizontal	θ_{H}	170	178	-	Degree	7
	Vertical	θ_{V}	170	178	-		
Effective	Horizontal	$\theta_{\text{GMA_H}}$		178	-	Degree	8
	Vertical	$\theta_{\text{GMA_V}}$		178	-		
Gray Scale				2.2			9



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Notes 1. Contrast Ratio(CR) is defined mathematically as : (By PR880)

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

It is measured at center point(Location P1)

2. Surface luminance(L_{WH})is luminance value at 1 point across the LCD surface 50cm from the surface with all pixels displaying white. For more information see FIG 5.

3. The variation in surface luminance , δ WHITE is defined as : (By PR880)

$$\delta_{\text{WHITE}} = \frac{\text{Minimum}(L_{P1}, L_{P2}, \dots, L_{P9})}{\text{Maximum}(L_{P1}, L_{P2}, \dots, L_{P9})} \times 100$$

Where L1 to L9 are the luminance with all pixels displaying white at 9 locations.
For more information see FIG 5.

4. Response time is the time required for the display to transition from black to white (Rise Time, Tr_R) and from white to black (Decay Time, Tr_D). For additional information see FIG 5.

5. Gray to gray response time is the time required for the display to transition from gray to gray. For additional information see Table 8. (By PR880)

6. Color shift is the angle at which the color difference is lower than 0.04.

For more information see FIG 6. (By EZ Contrast)

- Color difference ($\Delta u'v'$)

$$u' = \frac{4x}{-2x + 12y + 3} \quad v' = \frac{9y}{-2x + 12y + 3}$$

$$\Delta u'v' = \sqrt{(u'_1 - u'_2)^2 + (v'_1 - v'_2)^2} \quad \begin{array}{l} u'_1, v'_1 : u'v' \text{ value at viewing angle direction} \\ u'_2, v'_2 : u'v' \text{ value at front } (\theta=0) \end{array}$$

- Pattern size : 25% Box size

- Viewing angle direction of color shift : Horizontal, Vertical

7. Viewing angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface. For more information see FIG 7.
(By PR880)

8. Effective viewing angle is the angle at which the gamma shift of gray scale is lower than 0.3. For more information see FIG 8 and FIG 9.

9. Gray scale specification

Gamma Value is approximately 2.2. For more information see Table 9.

Product Specification

Measuring point for surface luminance & measuring point for luminance variation.

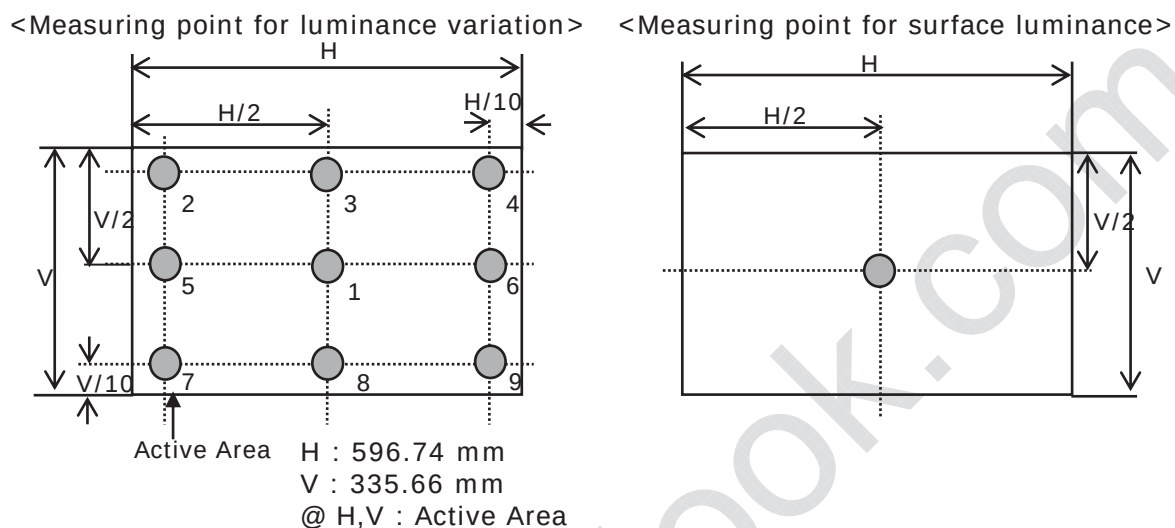


FIG. 5 Measure Point for Luminance

The response time is defined as the following figure and shall be measured by switching the input signal for “black” and “white”.

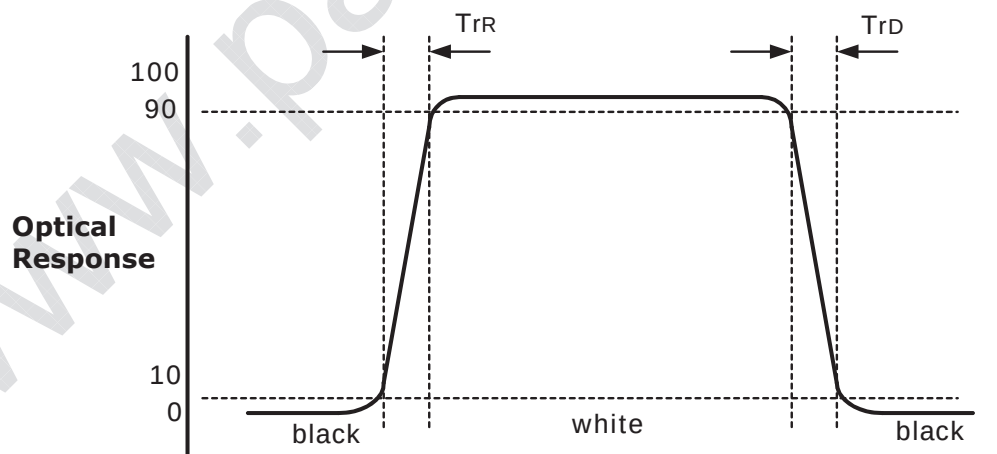


FIG. 6. Response Time



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Color shift is defined as the following test pattern and color.

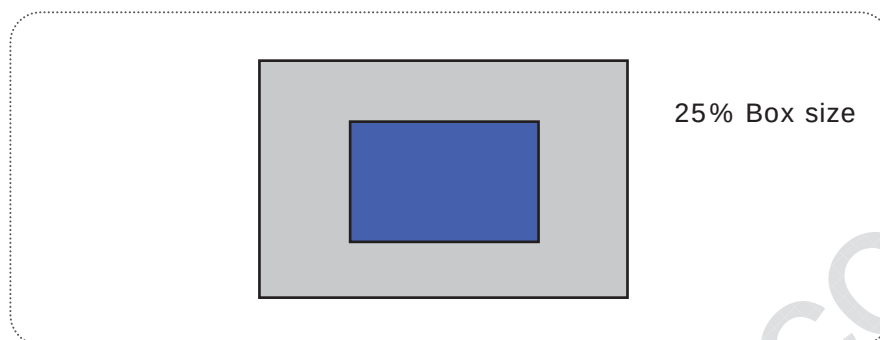


FIG. 7 Test Pattern

Average RGB values in Bruce RGB for Macbeth Chart

	Dark skin	Light skin	Blue sky	Foliage	Blue flower	Bluish green
R	98	206	85	77	129	114
G	56	142	112	102	118	199
B	45	123	161	46	185	178
	Orange	Purplish blue	Moderate red	Purple	Yellow green	Orange yellow
R	219	56	211	76	160	230
G	104	69	67	39	193	162
B	24	174	87	86	58	29
	Blue	Green	Red	Yellow	Magenta	cyan
R	26	72	197	241	207	35
G	32	148	27	212	62	126
B	145	65	37	36	151	172
	White	Neutral 8	Neutral 6.5	Neutral 5	Neutral 3.5	black
R	240	206	155	110	63	22
G	240	206	155	110	63	22
B	240	206	155	110	63	22

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Dimension of viewing angle range.

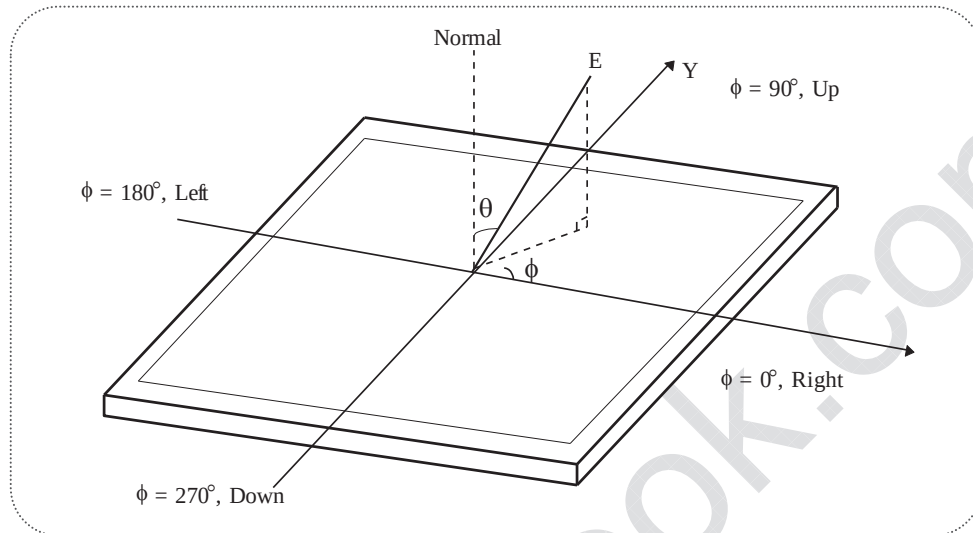


FIG. 8 Viewing angle

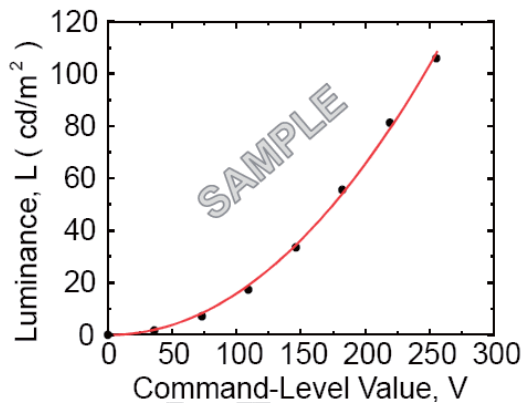


FIG. 9 Sample Luminance vs. gray scale
(using a 256 bit gray scale)

$$L = aV^r + L_b$$

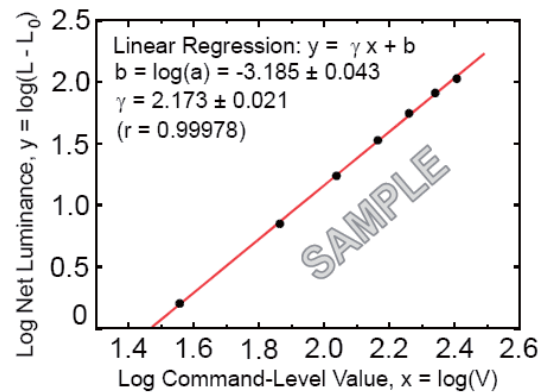


FIG. 10 Sample Log-log plot of
luminance vs. gray scale

$$\log(L - L_b) = r \log(V) + \log(a)$$

Here the Parameter a and γ relate the signal level V to the luminance L .

The GAMMA we calculate from the log-log representation (FIG. 10)

**LM270WQ1**
Liquid Crystal Display**Product Specification****Table 9. Gray Scale Specification**

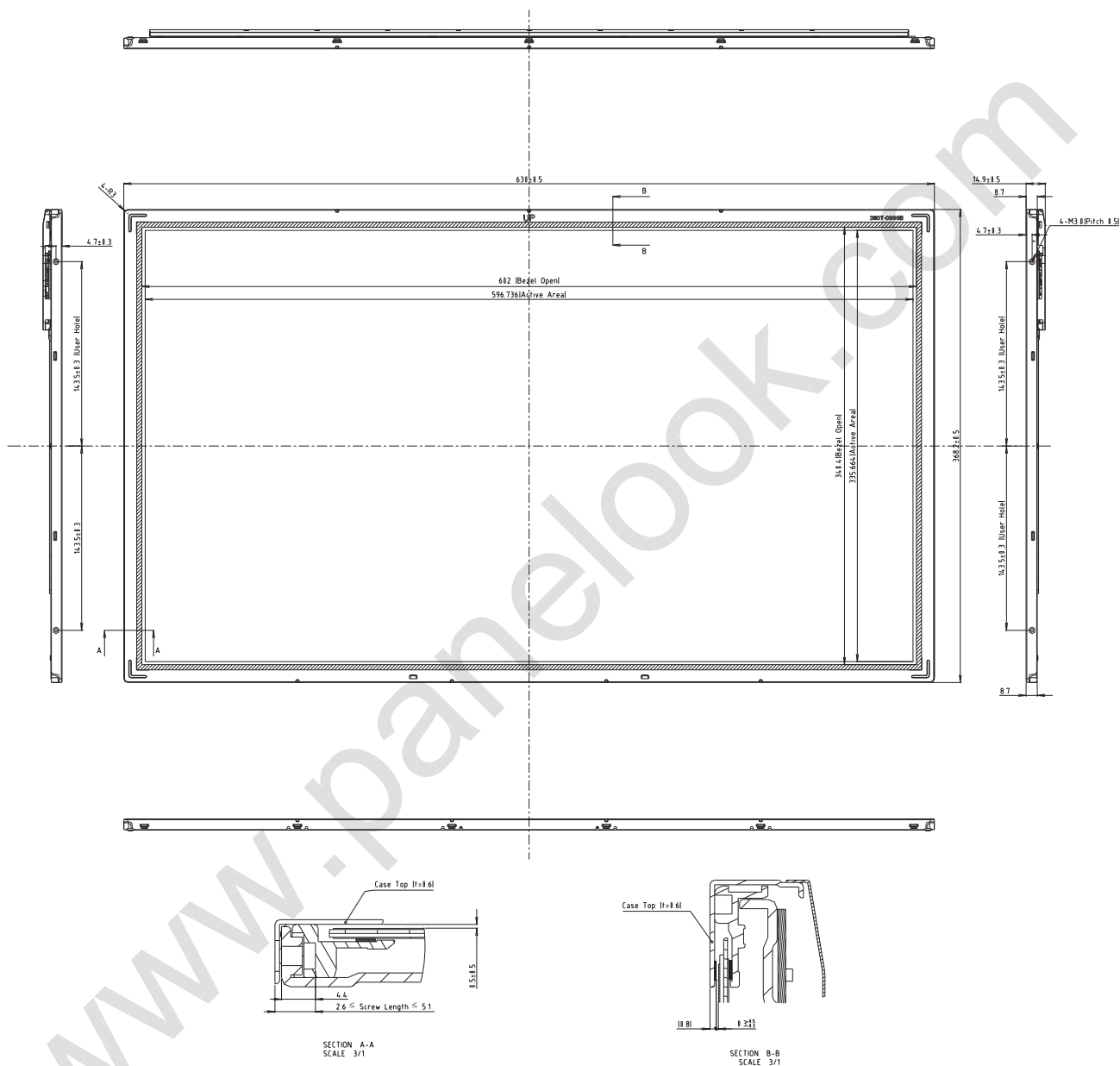
Gray Level	Relative Luminance [%] (Typ.)
0	0.10
15	0.30
31	1.08
47	2.50
63	4.71
79	7.70
95	11.52
111	16.18
127	21.72
143	28.15
159	35.51
175	43.81
191	53.07
207	63.30
223	74.52
239	86.75
255	100

**LM270WQ1**
Liquid Crystal Display**Product Specification****5. Mechanical Characteristics**

The contents provide general mechanical characteristics. In addition the figures in the next page are detailed mechanical drawing of the LCD.

Outline Dimension	Horizontal	630.0mm
	Vertical	368.2mm
	Depth	14.9mm
Bezel Area	Horizontal	602.0mm
	Vertical	340.4mm
Active Display Area	Horizontal	596.74mm
	Vertical	335.66mm
Weight	2,620 g (Typ.) / 2,750 g (Max.)	
Surface Treatment	Hard coating(3H) Anti-glare treatment of the front polarizer	

Notes : Please refer to a mechanic drawing in terms of tolerance at the next page.

**LM270WQ1**
Liquid Crystal Display**Product Specification****<FRONT VIEW>**

**LM270WQ1**
Liquid Crystal Display**Product Specification****6. Reliability**

Environment test condition

No	Test Item	Condition
1	High temperature storage test	Ta= 60°C 240h
2	Low temperature storage test	Ta= -20°C 240h
3	High temperature operation test	Ta= 50°C 50%RH 240h
4	Low temperature operation test	Ta= 0°C 240h
5	Vibration test (non-operating)	Wave form : random Vibration level : 1.0Grms Bandwidth : 10-300Hz Duration : X,Y,Z, 10 min One time each direction
6	Shock test (non-operating)	Shock level : 100G Waveform : half sine wave, 2ms Direction : $\pm X$, $\pm Y$, $\pm Z$ One time each direction
7	Altitude Operating Storage / Shipment	0 - 10,000 feet(3,048m) 0 - 40,000 feet(12,192m)

**LM270WQ1**
Liquid Crystal Display**Product Specification****7. International Standards****7-1. Safety**

- a) UL 60950-1, Underwriters Laboratories Inc.
Information Technology Equipment - Safety - Part 1 : General Requirements.
- b) CAN/CSA-C22.2 No. 60950-1-07, Canadian Standards Association.
Information Technology Equipment - Safety - Part 1 : General Requirements.
- c) EN 60950-1, European Committee for Electrotechnical Standardization (CENELEC).
Information Technology Equipment - Safety - Part 1 : General Requirements.
- d) IEC 60950-1, The International Electrotechnical Commission (IEC).
Information Technology Equipment - Safety - Part 1 : General Requirements

7-2. EMC

- a) ANSI C63.4-2003 "American National Standard for Methods of Measurement of Radio-Noise Emissions from Low-Voltage Electrical and Electronic Equipment in the Range of 9kHz to 40GHz."
American National Standards Institute (ANSI), 2003.
- b) C.I.S.P.R. Pub. 22. Limits and methods of measurement of radio interference characteristics of information technology equipment." International Special Committee on Radio Interference (C.I.S.P.R.), 2005.
- c) EN 55022 "Limits and methods of measurement of radio interference characteristics of information technology equipment." European Committee for Electrotechnical Standardization (CENELEC), 2006.

7-3. Environment

- a) RoHS, Directive 2011/65/EU of the European Parliament and of the council of 8 June 2011

**LM270WQ1**
Liquid Crystal Display**Product Specification****8. Packing****8-1. Designation of lot mark**

a) Lot mark

A	B	C	D	E	F	G	H	I	J	K	L	M
---	---	---	---	---	---	---	---	---	---	---	---	---

A,B,C : Size (Inch)

E : Month

D : Year

F ~ M : Serial No.

Note:

1. Year

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Mark	A	B	C	D	E	F	G	H	J	K

2. Month

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Mark	1	2	3	4	5	6	7	8	9	A	B	C

b) Location of lot mark

Serial No. is printed on the label. The label is attached to the backside of the LCD module.
This is subject to change without prior notice.

8-2. Packing Form

a) Package quantity in one box : 10ea

b) Box Size : 355mm X 700mm X 430mm



LM270WQ1 Liquid Crystal Display

Product Specification

9. Precautions

Please pay attention to the followings when you use this TFT LCD module.

9-1. Mounting Precautions

- (1) You must mount a module using holes arranged in four corners or four sides.
- (2) You should consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the Module. And the case on which a module is mounted should have sufficient strength so that external force is not transmitted directly to the module.
- (3) Please attach the surface transparent protective plate to the surface in order to protect the polarizer. Transparent protective plate should have sufficient strength in order to the resist external force.
- (4) You should adopt radiation structure to satisfy the temperature specification.
- (5) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the polarizer at high temperature and the latter causes circuit break by electro-chemical reaction.
- (6) Do not touch, push or rub the exposed polarizers with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of polarizer for bare hand or greasy cloth.
(Some cosmetics are detrimental to the polarizer.)
- (7) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach front / rear polarizers. Do not use acetone, toluene and alcohol because they cause chemical damage to the polarizer.
- (8) Wipe off saliva or water drops as soon as possible. Their long time contact with polarizer causes deformations and color fading.
- (9) Do not open the case because inside circuits do not have sufficient strength.
- (10) As The IPS panel is sensitive & slim, please recommend the metal frame of the system supports the panel by the double side-mount.

9-2. Operating precautions

- (1) The spike noise causes the mis-operation of circuits. It should be lower than following voltage : $V = \pm 200\text{mV}$ (Over and under shoot voltage)
- (2) Response time depends on the temperature. (In lower temperature, it becomes longer.)
- (3) Brightness depends on the temperature. (In Higher temperature, it becomes lower.)
And in lower temperature, response time (required time that brightness is stable after turned on) becomes longer.
- (4) Be careful for condensation at sudden temperature change. Condensation makes damage to polarizer or electrical contacted parts. And after fading condensation, smear or spot will occur.
- (5) When fixed patterns are displayed for a long time, remnant image is likely to occur.
- (6) Module has high frequency circuits. Sufficient suppression to the electromagnetic interference shall be done by system manufacturers. Grounding and shielding methods may be important to minimized the interference.
- (7) Please do not give any mechanical and/or acoustical impact to LCM. Otherwise, LCM can not be operated its full characteristics perfectly.
- (8) A screw which is fastened up the steels should be a machine screw (if not, it causes metal foreign material and deal LCM a fatal blow)
- (9) Please do not set LCD on its edge.
- (10) When LCMs are used for public display defects such as Yogore, image sticking can not be guarantee.
- (11) Partial darkness may happen during 3~5 minutes when LCM is operated initially in condition that luminance is under 40% at low temperature (under 5℃). This phenomenon which disappears naturally after 3~5 minutes is not a problem about reliability but LCD characteristic
- (12) LCMs cannot support "Interlaced Scan Method"

**LM270WQ1**
Liquid Crystal Display**Product Specification****9-3. Electrostatic discharge control**

Since a module is composed of electronic circuits, it is not strong to electrostatic discharge. Make certain that treatment persons are connected to ground through wrist band etc. And don't touch interface pin directly.

9-4. Precautions for strong light exposure

Strong light exposure causes degradation of polarizer and color filter.

9-5. Storage

When storing modules as spares for a long time, the following precautions are necessary.

- (1) Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- (2) The polarizer surface should not come in contact with any other object.
It is recommended that they be stored in the container in which they were shipped.

9-6. Handling precautions for protection film

- (1) The protection film is attached to the bezel with a small masking tape.
When the protection film is peeled off, static electricity is generated between the film and polarizer. This should be peeled off slowly and carefully by people who are electrically grounded and with well ion-blown equipment or in such a condition, etc.
- (2) When the module with protection film attached is stored for a long time, sometimes there remains a very small amount of glue still on the bezel after the protection film is peeled off.
- (3) You can remove the glue easily. When the glue remains on the bezel surface or its vestige is recognized, please wipe them off with absorbent cotton waste or other soft material like chamois soaked with normal-hexane.